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Roll No

ME-7004(2)-CBGS

B.E. VII Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Reliability Engineering

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1.
 - a) Explain with examples random variables.
 - b) Explain about different types of reliability analysis methods.
2.
 - a) Explain a few system failures with suitable examples.
 - b) Write salient features of reliability management.
3.
 - a) Three power supplier are configured in a standby redundant system with perfect switching. The failure rate for each of the power supplies is constant with a mean time between failures of 20,000 hours. What is the probability of the system failing in less than 1,00,000 hours?
 - b) A pump operates continuously with a mean time to fail of 200 hours that follows the exponential distribution. A second, identical pump is placed in standby redundancy, and the mean time to fail while the pump is inactive is 1,000 hours. The standby time to fail is also exponentially distributed. What is the mean time to fail for the system, and what is the system reliability at time = 300 hours?

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4.
 - a) Discuss about the application of probabilistic Risk Assessment in any industry.
 - b) Write short notes any two on the following.
 - i) Economic life
 - ii) Reliability Allocation
 - iii) Survival Curves
5.
 - a) Describe the stepwise procedure of constructing a fault tree with an example of your choice.
 - b) Discuss in detail various models used in the demonstration of maintainability of a unit Illustrate the use of repair time distributions.
6.
 - a) State and explain following components of reliability-probability, intended function time and operating conditions.
 - b) What are the various causes of failures of components?
7.
 - a) Draw a graph which depicts failure rate Vs time and analyse that graph.
 - b) Write notes on series, parallel and mixed system and give suitable examples.
8. Write short notes on
 - a) Laws of probability
 - b) Monte Carlo simulation.
